Quang Huy Dao

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Education

Hanoi University of Science and Technology (HUST) Hanoi, Vietnam

School of Electric – Electrical Engineering Talent Program of Control Engineering and Automation *Sep 2021 – August 2025*

GPA: 3.83/4.00

Experience

VinMotion Inc. California, USA

Robotics Engineer *October 2026 – Present*

- Whole-body control for humanoid robots.
- General motion tracking and perceptive whole-body control.
- Behavioral foundation models for humanoid control.

VinMotion Hanoi, Vietnam

Robotics Engineer *April 2025 – September 2026*

- Developed and trained Reinforcement Learning (RL) and Imitation Learning (IL) policies for locomotion and imitation tasks on physical humanoid robots.
- Implemented Whole-Body Control (WBC) frameworks for humanoid robots, coordinating full-body dynamics to achieve stable.
- Designed and deployed Model Predictive Control (MPC) strategies for humanoid robot locomotion.
- Developed state estimation pipelines for humanoid robots using Invariant extended Kalman filter (InEKF).

School of Electrical Engineering, HUST Hanoi, Vietnam

Scientific research *May 2022 – August 2025*

- Research on Trajectory Optimization for Robots. A high-order control barrier function is used to enhance obstacle avoidance ability.
- Research on Inverse Reinforcement Learning for Intelligent Systems.
- Designing an Off-policy Reinforcement Learning algorithm for multi-agent systems.
- Apply RL to design algorithms for the Nash equilibrium problem to minimize the impact of noise on smart systems.

Awards & Achievements

Employee of the Year, VinMotion: Selected as the outstanding employee of the year. (2025)

Second prize in the Vietnam Mathematical Olympiad for Students University: Mathematics competition for undergraduate students in Vietnam. (April 2022)

Second prize in the Mathematical provincial round in the Vietnam Student Olympiad: Mathematics competition for high school students. (Dec 2020)

Certificate of Merit in the Provincial Scientific Research Student Award: Scientific research competition for high school students. (Dec 2019)

Publication

Chuong Nguyen*, **Huy Dao***, Loc Pham*, Truong Nguyen and Quan Nguyen, "Robust and Optimized Aggressive Standing-Up Maneuvers for Humanoid Robots across Diverse Terrains" (accepted at IEEE-RAS International Conference on Humanoid Robots, Humanoids 2026).

Anh Duc Tran*, Anh-Quang Vu*, Loc Pham, Trong Hieu Nguyen, Chuong Nguyen Le, **Huy Dao**, Hoang Vu Dao, Van-Truong Nguyen, Chuong Nguyen, Tuyen P. Le and Quan Nguyen, "Ukemi-SafeFall: An Ukemi-inspired Injury-Aware Falling and Recovery" (accepted at IEEE-RAS International Conference on Humanoid Robots, Humanoids 2026).

Dai-Nhan Duong*, **Huy Dao***, Khoa Vu, Hieu Luong, Thuong Tran, An Le, Tuyen P. Le, Vu Dao, Truong Nguyen and Quan Nguyen, "X-BFM: a Behavioral Foundation Model for Extreme Humanoid Control via World-Model Conditioning and Decoupled Recovery Priors" (under review at ICRA).

K. Thai, **H. Dao** and N. T.-T. Vu, "Stochastic Integral Reinforcement Learning for Distributed H_∞ Optimal Formation Tracking of Multi-Agent Wheeled Mobile Robots" (under review at the International Journal of Robust and Nonlinear Control).

D. C. Nguyen, **Q. H. Dao** and P. N. Dao, "A model free approach for continuous-time optimal tracking control with unknown user-define cost and constrained control input via advantage function" (paper, under review at Aerospace Science and Technology).

P. N. Dao and **Q. H. Dao**, "H_∞ Tracking Control for Linear Discrete-Time Systems using On/Off Policy Q-Learning Algorithms." (accepted at Chaos, Solitons and Fractals).

T. T. Cao, **Q. H. Dao** and P. N. Dao, "Q-Learning Algorithm for F16 Aircraft: An approach to Discrete-Time Systems." 2023 12th International Conference on Control, Automation and Information Sciences (ICCAIS), Hanoi, Vietnam, 2023, pp. 1-4. (accepted).

Projects

X-BFM | *Project page*

- Project page for a behavioral foundation model for extreme humanoid control, using world-model conditioning and decoupled recovery priors.

Aggressive Standing-Up (TO-DRL) | *Project page*

- Project page for robust and optimized aggressive standing-up maneuvers for humanoid robots across diverse terrains, conducted at VinMotion (Humanoids 2026).

Ukemi-SafeFall | *Project page*

- Project page for Ukemi-inspired injury-aware falling and recovery, letting a humanoid fall while limiting impact and stand back up (Humanoids 2026).

Implementation of the iMPC-DHOCBF Algorithm | *GitHub*

- DHOCBFiMPC4DyEnv uses iterative Model Predictive Control (iMPC) with a Quadratic Programming (QP) framework and can solve problems using QP solvers.

Skills

Languages: English(B2)

Programing Languages: Matlab, Python, C/C++ ,Julia

Technologies: Git, Eigen, ROS 2 (Robot Operating System 2), Torch, JAX, OCS2 (Optimal Control of Switched System toolbox) , MuJoCo, IsaacGym, IsaacLab, Mjlab and various optimization libraries.